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Democratic Economic Cell Network

From the national foundation to the planetary horizon — Patch v4.4.3: use-based ownership & defense of the commons.

Version 5.0.1 — corrected edition

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v4.4.3 was renamed v5 owing to a new integrated theoretical core; v5.0.1 corrects the versioning, the Appendix A formulas, and the formatting.

Patch v4.4.3 application note & v5.0.1 corrective

v5.0.1 corrective — June 2026

This edition applies the RCED V5 corrective patch and resolves the versioning, formula and threshold inconsistencies. Corrections made: **(1)** allocation constraint F1 written $\alpha + \beta + \gamma + \delta + \varepsilon = 1$ (ε replaces the erroneous “x”); **(2)** the federal contribution ε trigger threshold set at **2% of GDP** (not 21%); **(3)** syntactic correction of the energy balance F4; **(4)** terminological consolidation of the Commons Use Levy and of the Commons Guard's role; **(5)** rebuilding of the layout (table of contents, lists, Appendix A, glossary).

The patch file indicates a source version 4.4.2 and a target 4.4.3. As v4.4.2 is unavailable, the patch is applied directly to the v4.4.1 base. No v4.4.1 content is removed. The numbering conflict has been resolved: the Ecological Loop remains *Layer 12*, and the defense bodies introduced by the patch are added as distinct layers (Commons Guard as internal security, then Layers 15 Territorial Defense, 16 Immunity, 17 Verification).

These additions belong to Part II (aspirational horizon, not deliverable within 20 years) under a cautious framing: consent, civilian control, interposition and non-imposition, last resort, and prior human verification.

Table of Contents

1. Executive Summary

The RCED proposes a radical yet pragmatic transformation of the Swiss economic system. Version v4.4.3 preserves the entirety of the foundation's achievements (v4.3.2, then Layers 13-14 of v4.4.1): transparency via blockchain, cooperative governance, democratic control, common infrastructures, the ecological loop, and the Decarbonation Fund. It applies the v4.4.3 patch on top, which carries out a major paradigm shift and formalizes the defense of the commons.

The v4.4.3 patch can be summed up in four pillars: **(1)** capitalist private property gives way to use-based ownership; **(2)** classic redistributive taxation is replaced by a Commons Use Levy; **(3)** the model moves from a mere fiscal-transition tool to a structure of direct social production; **(4)** the defense of the commons is rigorously structured (Commons Guard, territorial defense, immunity, verification).

Key results — Reference Scenario (Year 20)

The Reference Scenario projections for Part I demonstrate exceptional macroeconomic and ecological performance without money creation, while maintaining a strict gain-allocation constraint.

+180.4% **GDP Index** Base 2026 = 100 (+180.4 pts)	**72 / 100** **Gross National Happiness** Progress of +14 points	**0.185** **Gini Coefficient** Sharp reduction (−0.135)
28.5 h **Working hours / week** Time freed (−12.5 h)	**−68%** **Net emissions** Net-zero 2050 trajectory (LCI)	**19.7%** **Sovereign Fund surplus** Expressed as % of GDP

Complementary indicator	Y20 value	Impact / Justification
Networks placed in common	3 networks	Rail, energy and digital, open and rent-free.
Final energy (Y20 Index)	77.4	22.6% drop in final consumption.
Overbuild required (Layer 14)	1.25×	Optimized vs ~2× required without a wide interconnected network.

2. Evolution from v4.3.2 to v4.4.3

Version v4.4.x invalidates none of the corrections of version v4.3.2 but extends them. The foundation (Layers 0 to 12) and the eight numerical-consistency checks remain unchanged. The model is anchored in an incremental logic drawing on existing precedents (Swiss Federal Railways CFF, Swissgrid, civil service, EU energy communities) while introducing foundations from Karl Marx (property), Elinor Ostrom (commons and graduated sanctions) and radical subsidiarity.

Element (base)	Evolution with patch v4.4.3	Justification / Impact
γ reallocation (non-automatable sector)	Layer 13: qualifying citizen service (renovation, energy, care).	Provides a concrete operator for the 3-to-5× pace required for a 100% renewable system.
Decarbonation Fund (δ)	Layer 13: thermal storage (sand) + local resources.	Removes dependence on critical minerals; seasonal storage; lowers final consumption.

National commons (Layer 11)	Layer 14: federated energy cells, energy subsidiarity.	Smooths intermittency across space, reduces overbuild from ~2× → 1.25×.
International-scale limit of electricity	Part II: Earth Federation Project, sequenced after the foundation matures.	Treats the political blind spot as a horizon distinct from the deliverable layers.
Capitalist private property & redistributive taxation	Institutionalized use-based ownership and Commons Use Levy (Layer 2).	The asset is tied to its effective use. We no longer redistribute after the fact; we bill at the source.
Financial-wealth register (Layer 1)	Cadastre of uses (energy, land, digital).	The audit bears on the real use of resources, consistent with use-based ownership.
Collective security (§12, horizon)	Commons Guard + Layers 15-17 (defense, immunity, verification).	Graduated defense of the commons: enclosure prevention, automatic immunity, human validation.

3. Paradigm shift (patch v4.4.3)

Purpose: the model no longer presents itself as a mere fiscal-transition tool, but as a structure of direct social production: economic coordination runs through the use of the commons rather than through accumulation and late redistribution.

3.1 From private property to use-based ownership

The patch abolishes private property as defined by capitalism (the title of capital as a right to exclude and to collect a rent). It substitutes use-based ownership: an asset is constitutionally tied to whoever effectively uses it, according to the rules of the commons, rather than to a detachable title. Inspired by Marx read through Ostrom, this principle does not forbid possessing what one uses (housing, tools, a business), but removes the possibility of capturing a rent on an unused asset that deprives others of its use.

3.2 From redistributive taxation to the commons use levy

The redistributive taxation system (Piketty / Zucman / Saez) is entirely replaced by a Commons Use Levy. Instead of levying ex-post on incomes and wealth in order to redistribute, the model bills at the source the use of common resources (energy, land, data, register capacity). The contribution becomes a condition of access to the commons, falling within the Ostromian logic of access-dues. Layer 0 (diagnosis of inequalities) remains relevant as a starting observation, but the instrument shifts toward the use levy.

4. The full layer stack (foundation + patch)

The overall structure integrates the historical layers of the initial foundation reinterpreted by patch v4.4.3, completed by the ecological-transition innovations (13-14) and the security ones (15-17).

#	Element / Layer	Institutional function & mechanism
0	**Diagnosis**	Observation of inequalities (analytical starting point, no longer a mechanism of action).
1	**Register of uses**	Cadastre of energy, land, digital and carbon uses on Blockchain (patch v4.4.3).
2	**Use levy**	Replaces redistributive taxation: source billing of the use of the commons (patch v4.4.3).
3	**Capitalization**	Citizen sovereign fund investing directly in the real economy.
4a	**Production**	Workers' cooperatives.
4b	**Consumption**	Consumers' cooperatives and democratic planning of supply.
5	**Health / Time**	Solidarity insurance and introduction of the reduced working week.
6	**Coordination**	Federated cells, using the free market merely as a transmission signal.
7	**Governance**	Swiss model based on strict subsidiarity, from municipalities to the Confederation.
8	**Knowledge**	Official training (SERI / OrTra) with credentialing reconversion schemes.
9	**Propagation**	0% investment enabling the model to spread and create cells internationally.
10	**Time**	Indexation of labor: required time falls in proportion to automation.
11	**National commons**	Rail, energy and digital in open access to purge the economy of rents.
12	**Ecological loop**	Strict carbon budget + Decarbonation Fund (δ) — net-zero LCI 2050 trajectory.
13	**Citizen mobilization**	Citizen service + sober thermal storage + structural relocalization (§ 5).
14	**Energy cells**	Nested energy sharing and networked energy subsidiarity (§ 6).
I.S.	**Commons Guard**	Internal security: enclosure prevention, technical audit, administrative interposition.
15	**Territorial defense**	Subsidiary and decentralized external defense, financed by the commons dividend.
16	**Immunity**	“Broadcast” protocol: automatic and reversible isolation of offending cells.
17	**Verification**	Decentralized media consensus (human-in-the-loop) to validate genuine aggression.

4.1 Gain-sharing mechanism (recap)

For each year *t*, the annual automation rate $\Delta P(t) = r$ (estimated between 2% and 6%) is rigorously allocated according to the following allocation constraint:

ALLOCATION CONSTRAINT

$$\alpha + \beta + \gamma + \delta\ (+\ \varepsilon) = 1$$

The default coefficients of the operational foundation are defined as follows:

- **α = 45%** → feeds the reduction of working time ($\alpha \times \Delta P(t)$).

- **β = 35%** → feeds the citizen Sovereign Fund ($\beta \times \Delta P(t)$).
- **γ = 10%** → allocated to sectoral reallocation and training / Layer 13 ($\gamma \times \Delta P(t)$).
- **δ = 10%** → allocated to the Decarbonation Fund / Layer 12 ($\delta \times \Delta P(t)$).
- **ε** (federal contribution / Part II) → conditional, activated only if the Sovereign Fund surplus exceeds **2% of GDP**.

Golden rule: 100% of automation gains remain allocated, with no money creation whatsoever.

Democratic governance note

The annual arbitration and modulation of the macroeconomic coefficients (α, β, γ, δ, ε) belong to no fixed algorithm nor to technocratic management. The allocation of this budget is sovereignly deliberated and validated by a Citizens' Assembly, ensuring the continuous alignment of production and of working-time reduction with the real needs expressed by the population.

5. Layer 13 — Citizen mobilization of the transition

The deployment pace required to replace fossil fuels with renewable energy is 3 to 5 times higher than the current pace. The conventional labor market can supply this workforce only with problematic slowness. Layer 13 articulates three major levers to accelerate this transition while lowering final energy consumption without degrading well-being:

- **Citizen service of the transition** — redirecting an existing institution toward critical worksites (§ 5.1).
- **Sober thermal storage** — sand batteries for seasonal heat storage (§ 5.2).
- **Renovation & relocalization** — structural reduction of final-energy demand (§ 5.3).

5.1 Citizen service of the transition

The principle is to redirect an existing, consensual Swiss institution (compulsory military/civil service) rather than to build an unprecedented form of coercion. Fit young adults perform a qualifying service, supervised by professionals, channeled toward the critical worksites of the transition: thermal renovation of buildings, installation of renewable infrastructure and heat pumps, ecological restoration, and support to the care sector.

- **Articulation with Layer 8 (knowledge):** the pathway leads to an official certification or diploma; mobilization and reconversion merge into a single mechanism.
- **Articulation with the non-automatable sector:** the service feeds the ecology, care and education sectors, directly nourishing Gross National Happiness (GNH).

Imperative safeguards to integrate

1. **Individual freedom:** the obligation remains a constraint; it is justified only if the service stays short, qualifying, and guarantees a real choice of field.
2. **Productivity:** an untrained conscript is less efficient than a skilled tradesperson. The service must complement skilled artisans rather than replace them, requiring dense professional supervision.
3. **Accounting:** the opportunity cost must be rigorously accounted for as a transfer of labor (not as free work) in the consolidated balance sheet.

5.2 Sober thermal storage — sand batteries

A fundamental technical distinction is required: one does not store electricity, one stores heat. Sand is brought to a temperature of 500-600 °C using surplus renewable electricity. It then releases this heat over days or months via urban heat networks (district heating).

What sand storage solves	What must be clear-sightedly acknowledged
Mineral independence: no recourse to lithium, cobalt, nickel or rare earths. Uses only sand, steel and a heating element. A local, abundant and neutral resource.	**Poor electrical efficiency:** heavy losses when converting heat back into electricity. Unbeatable for heating, but unsuited to mobility.
Seasonal storage: able to transfer energy from summer to winter, which classic electrochemical storage cannot do at scale.	**Heavy infrastructure:** presupposes the existence or deployment of vast heat networks, which is an infrastructure project in itself.
Peak management: efficiently absorbs renewable overproduction peaks instead of having to curtail them.	**Complementarity:** it is an indispensable complement, not a substitute, to electrochemical storage (reserved for mobility and electronics).

5.3 Renovation, relocalization and energy balance

In a country with a temperate/cold climate, building heating accounts for nearly 45% of final energy consumption. A deep thermal renovation of the envelope cuts heat demand by 50 to 80%, while a heat pump (HP) with a COP of 3 to 4 drastically divides the primary energy required.

Transition item	Measured effect on final consumption
Building heating (renovation + HP)	↓↓↓ Dominant and priority lever.
Freight transport (relocalization)	↓ Net reduction of physical flows.
Seasonal storage (sand)	Neutral — secures 100% renewable thermal supply.
Specific electricity (global electrification)	→ or ↑ Stable to slightly rising consumption.
Embodied energy of the transition (launching worksites)	↑ Temporary rise, amortized thereafter.
Net balance (Y20 Index = 77.4)	**Overall 22.6% drop** in final consumption, at constant well-being.

6. Layer 14 — Federated energy cells

The residual blind spot of the transition concerns the specific intermittency of electricity. Layer 14 addresses it by deploying network sharing, acting in perfect synergy with storage solutions. The model articulates two physical dimensions: **time** (moving energy from one moment to another via local storage) and **space** (moving energy from one geographic point to another via the meshed network to smooth instantaneous production).

Scale	What space-sharing solves	What it cannot solve on its own
Neighborhood	Fine demand diversity, reduced line losses, optimized self-consumption.	Day/night cycle, local weather hazards (zone production stays correlated).
National	Smoothing of microclimates and regional weather differences.	Day/night cycle (Switzerland has too narrow a longitudinal extent).
Continental	Strong weather compensation, exploiting time-zone offsets and wind/solar anti-correlation.	Deep, persistent troughs of winter nights (Dunkelflaute phenomena).
Global	Complete resolution of the day/night cycle (it is continuously daytime on one face of the Earth).	Physically robust, but reopens the immense challenge of political governance (see Part II).

6.3 Guiding principle: energy subsidiarity

The network operates on a fractal, nested architecture: local self-sufficiency is maximized first. Each higher tier intervenes only to regulate the energy residual not absorbable by the lower tier. Territorial autonomy is the inviolable rule, global interconnection is the safety net (and never the reverse). One does not depend on the large grid; one is connected to it as an insurance.

Impact on oversizing: thanks to this wide network and the meshing of Layer 14, the overbuild index (required oversizing of production infrastructure) collapses from ~2× (autonomous island system) to only **1.25×**, massively saving embodied energy and critical minerals.

Technical hookup: Layer 1 (Blockchain) transparently traces energy flows and peer-to-peer reciprocity credits between cells. Real regulatory precedents support this approach: self-consumption groupings (RCP/ZEV) in Switzerland and the citizen energy communities promoted by the European Union.

Note on the dual promotion trajectory

The long-term viability of the model rests on its dissemination. A symmetrical effort of cultural facilitation and diplomacy is required: internally (deployment of pilot portfolios in the cantons, local carbon accounting) and internationally (citizen and academic diplomacy aiming to standardize cell-based accounting and to launch Part II).

Part II — Earth Federation Project (aspirational horizon)

Status: a multigenerational, aspirational project, not deliverable within a 20-year horizon. Its activation remains rigorously conditioned on the full maturity of Part I (defined by the network's propagation to a critical mass of countries).

8. Status, sequencing and nature of the problem

The Earth Federation differs fundamentally in nature from Layers 13 and 14. The latter solved technical bottlenecks through engineering. The Earth Federation claims to fill the political blind spot, namely global coordination. Yet sovereignty does not vanish through a mere design trick. The two projects are therefore strictly sequenced: each phase must prove its robustness before the next can be contemplated.

10. Equalization through the commons dividend

To be realistic and politically tractable, global equalization is not organized as a direct transfer of wealth between rich and poor continents, but is backed by a dividend from the planetary commons themselves. The atmosphere (via the single carbon register), the international energy backbone, Earth orbit, the oceans and critical mineral deposits are defined as global commons.

The use levies collected at the source on these resources feed a Sovereign Fund transposed to the scale of the Earth. Redistribution operates on these centralized receipts: the paradigm is no longer “the rich pay for the poor,” but “the planetary commons pay their dividend and distribute it according to need.”

11. Financing: 0% investment → commons → dividend

The global financial engine reuses the propagation mechanism of Layer 9, inspired by the great historical cooperative federations (Migros, Coop, Raiffeisen, Mondragon) where members' surpluses finance the umbrella structures.

- **The investment:** the surplus of national economic cells is channeled, via the 0%-loan logic, toward building the infrastructures (continental interconnection cables, registers, environmental restoration).
- **The equalization:** once operational, these common infrastructures generate access levies. This dividend then finances structural redistribution.

Collecting without an army (access-dues): following Ostrom's theses, the global contribution is not based on an invasive customs or military levy, but takes the form of an unavoidable condition of access to the network of commons. To take part in exchanges, to use the energy infrastructure or the carbon register, a cell must be up to date with its access-dues. One does not coerce by force; one disconnects technical access.

14. Collective security: from the non-violent lever to the interposition corps

This tier is the most sensitive and most tightly framed component of Part II. Conceived as a last resort, it excludes any permanent centralized army and relies on two graduated instruments:

Instrument	Nature of the device & mechanism	Resort
Non-violent lever	Exclusion from access to the commons (carbon register, unified market, energy interconnection). One does not extract wealth; one technically disconnects the cell.	**First resort.** Mild and economical.
Consented interposition corps	Physical deployment between belligerents to freeze hostilities and prevent the resumption of violence. Strict Blue-Helmet logic, never assault troops.	**Last resort.** Heavy and exceptional.

Absolute red lines and security architecture

- **Interposition, not imposition:** the federal corps does not deploy against one camp to subdue or occupy it. It does not govern and levies no tax.
- **Prior consent:** deployment is imperatively by invitation or consented mandate; absolute prohibition on entering by force into a non-consenting sovereign State.
- **The enforcement dilemma:** without direct coercive power, the global register risks being no more than a dashboard; endowed with a power of imposition, it would drift toward a centralized, authoritarian world-State — precisely the dangerous structure the RCED model fights. The Ostromian confederal approach with enumerated competences serves as a firewall.

Appendix A — Mathematical formulas

F1 — Extended macroeconomic allocation constraint

$$\alpha + \beta + \gamma + \delta \ (+ \ \varepsilon) = 1$$

Where $\varepsilon \geq 0$ represents the federal contribution (Part II), capped and strictly null if the Sovereign Fund surplus is $\leq$ 2% of GDP.

F2 — Heat demand of the renovated building & HP consumption

$$Q_{heat}(t) = Q_{heat}(0) \times (1 - \tau_{reno})$$

$$E_{elec} = Q_{heat} \,/\, COP$$

Target parameters: reduction rate $\tau_{reno} \approx 0.5$ to 0.8 | coefficient of performance COP ≈ 3 to 4.

F3 — Seasonal thermal storage (sand)

$$S_{th}(t) = S_{th}(t-1) + \eta_{charge} \times E_{surplus}(t) - Q_{released}(t) - \text{losses}(t)$$

Cycle optimized from thermal to thermal for the absorption of instantaneous renewable surpluses.

F4 — Energy balance of the cell (Layer 14)

$$\text{Self-consumption} = \min(\ \text{Prod}_{local}\ ,\ \text{Cons}_{local}\ )$$

$$\text{Shared-surplus}(t) = \Sigma \max(\ \text{Prod} - \text{Cons}\ ,\ 0\ ) \rightarrow \text{deficit cells}$$

Oversizing target (overbuild target v4.4): 1.25× (versus ~2× without a meshed network).

F5 — Cascading federal financing (Part II)

$$\text{Invest}_{commons}(t) = \varepsilon \times \Delta P(t) \times GDP(t)$$

$$\text{Dividend}_{commons}(t) = \text{price}_{carbon} \cdot E_{taxed} + \text{mineral}_{royalties} + \text{levies}$$

$$\text{Equalization}(t) = \text{distribution}(\ \text{Dividend}_{commons}\ )$$

Appendix B — Glossary (new patch v4.4.3 entries)

Term	Definition and systemic role
Use-based ownership	Legal framework instituting the attachment of an asset to its effective user according to the charters of the commons. Removes the validity of detachable capital titles enabling passive rent extraction (a rereading of Marx through Ostrom).
Use levy	Mechanism of source billing for the use of common resources, fully replacing the former ex-post redistributive taxation schemes (Layer 2).
Commons Guard	Civil and administrative internal-security body tasked with auditing the register (Layer 1), alerting the assemblies, and pronouncing administrative interpositions (access exclusions) to block enclosure attempts.
Immunity (broadcast)	Automated technical protocol allowing the reversible and immediate isolation of an economic cell that fails to respect the common rules. A variant of the Ostromian graduated sanction.
Verification	Decentralized media-consensus device mandatorily including human validation (human-in-the-loop) to certify the existence of a physical aggression or a breach of the commons before any collective response.
Interposition corps	International peacekeeping force made of a thin core of professionals complemented by a reservoir of volunteer citizens. Positions itself strictly between belligerents to block violence.
Citizens' Assembly	Sovereign deliberative body tasked with annually modulating and validating the allocation of the macroeconomic parameters (α, β, γ, δ, ε) outside any technocratic automatism.

Appendix C — Sources and bibliographic references

- **Ostrom, E. (1990).** *Governing the Commons.* — General theory of the commons, access-dues, the architecture of graduated sanctions, and the defense of shared-governance structures.
- **Marx, K.** *Capital / Manifesto.* — Radical critique and principles for abolishing capitalist private property, transcribed here under the operational concept of use-based ownership.
- **Piketty, T. (2013).** *Capital in the Twenty-First Century.* — Modeling and databases on the historical concentration of capital, serving as the starting point for the diagnosis of inequalities (Layer 0).
- **Saez, E. & Zucman, G. (2019).** *The Triumph of Injustice.* — Empirical analyses of contemporary tax systems and redistribution dynamics.
- **Polar Night Energy (2022).** — First industrial and commercial high-temperature sand thermal-storage unit deployed in Finland (technical reference for Layer 13).
- **Cooperative models (Mondragon, Migros, Raiffeisen).** — References of federal and confederal structures financing their development and their internal solidarity mechanisms by pooling activity surpluses.
- **RCP/ZEV regulation (Switzerland).** — Legal framework for self-consumption groupings, serving as an empirical deployment model for the Layer 14 energy cells.